

Smart Automated Exhaust Fan for Heat-Intensive Environments

Components List

S.No	Component	Quantity
1	ESP32 Development Board	1
2	DHT22 Temperature & Humidity Sensor	1
3	Zero Cross Detector (ZCD) Circuit	1
4	MOC3021 Optocoupler	1
5	BT136 TRIAC	1
6	Bridge Rectifier	1
7	Step-Down Transformer (230V AC to 12V AC)	1
8	Voltage Regulator (7805/LM1117)	1
9	Exhaust Fan (AC)	1
10	Resistors (220Ω, 330Ω, 1kΩ, 10kΩ)	As Required
11	Capacitors	As Required
12	Diodes (1N4007)	As Required
13	PCB / Breadboard	1
14	Jumper Wires	As Required
15	Power Supply Unit	1
16	IoT Dashboard (Blynk/ThingSpeak)	1